



The Health Benefits of Cycling

Cycling for health and fitness

It is widely acknowledged that cycling is one of the best ways for people to achieve good health and fitness. People who cycle regularly live longer than those who do not and lead healthier lives [3]. This simple fact means that cyclists cannot possibly be more vulnerable than the population at large to life-threatening injuries such as head injury.

Indeed, cycling regularly to work (and, by extension, to school and on other regular journeys) has been shown to be the most effective thing an individual can do to improve health and increase longevity, and this applies even to people who are already active in sport and other physical activities [4]. Cyclists typically have a level of fitness equivalent to being 10 years younger [5]. A UK parliamentary health committee has noted:

“If the Government were to achieve its target of trebling cycling in the period 2000-2010 ... that might achieve more in the fight against obesity than any individual measure we recommend within this report.” [6]

The health benefits of cycling outweigh the risks [7], perhaps by 20 to 1 according to one estimate [8]. Safe though cycling is by comparison with other common activities, it has been shown that cycling becomes safer still the more people who do it - for a doubling in cycle use, risk decreases by a third [9] [10].

However, the converse of the above is also true. If fewer people cycle, perhaps because they have been deterred by cycle helmet laws or the exaggerated claims sometimes made to promote helmet wearing, then the health loss to both individuals and society is considerable, whilst those who continue to cycle will be at greater risk. It is likely that few people who quit cycling recover the loss of health benefits through other activity.

Health benefits

Below are summarised some of the health benefits of cycling, for the individual and society as a whole. Many of the benefits are not unique to cycling but are a consequence of moderate physical activity. In several cases, however, cycling enables that benefit to be achieved more easily, more widely or more effectively.

This should not be regarded as an authoritative treatment of the subject, relying as it does on the primary sources [1] [2] referenced below. Those sources provide much more detailed information and full references.

Immune system

The immune system serves to protect the body from infections and diverse systemic diseases. Studies show that moderate activity, such as cycling, strengthens the immune system and thus contributes to a healthy life. It can also increase activity against tumour cells, assisting the prevention of related illnesses.

Muscles

Humans have several hundred muscles, which have to be used regularly in order to maintain fitness and health. A week of inactivity reduces the strength of the muscular system by up to 50% and can harm them long-term. This is particularly true for older people as aging causes muscles to shrink.

During cycling, most of the body's muscles are activated. The leg muscles are responsible for the pedalling movement; the abdomen and back muscles stabilise the body on the cycle and cushion external influences; and the shoulder-arm muscular system supports the body at the handlebars. All this trains and tightens up the muscular system, making it stronger and able to function efficiently.

The skeletal system

This is the body's supporting framework, held together by muscles, tendons and ligaments. Exercise and the



accompanying strain on the skeletal system strengthens it and increases mobility. Cycling has a positive effect on bone density and strength. Moreover, a muscular system strengthened by cycling supports and protects the skeletal system.

Spinal diseases and back pains

Posture when cycling is optimum, and the cyclic movement of the legs stimulates muscles in the lower back, where slipped discs are most likely to occur. In this way the spine is strengthened and secured against external stresses. In particular cycling can stimulate the small muscles of the vertebrae which are difficult to affect through other exercise. This can help reduce the likelihood of back pains and other problems.

Joint protection

Cycling is especially good at protecting and feeding cartilages as the support given by the bicycle means that the forces that act as a result of body weight are significantly reduced. The circular movement of cycling assists the transport of energy and other metabolic produces to the cartilages, reducing the likelihood of arthrosis.

Balance and equilibrium

Physical activity serves as a regulator to relieve the stress that is common in current lifestyles. It produces the balance between exertion and relaxation which is so important for the body's inner equilibrium. Cycling is especially ideal for this process, countering stress in two ways: by satisfying the need for activity where people lack movement or exercise; and by balancing out increased strain, particularly mental and emotional.

Mental stress

Cycling has a considerable relaxing effect due to its uniform, cyclic movement which stabilises the physical and emotional functions of the body. This counteracts anxiety, depression and other psychological problems. The exercise also controls hormonal balance.

Oxygen and circulation

Oxygen is vital for all biological organisms and the basic prerequisite for the respiratory processes of humans. Respiration is often impaired by adiposity and lack of exercise. Among other things, regular physical activity strengthens the respiratory muscles, which leads to improved ventilation of the lungs and thus has a positive effect on oxygen exchange. Enormous positive health effects can be achieved in energy uptake and processing, through moderate cycling.

Heart and cardiovascular diseases

The heart is one of the most important organisms for a healthy life but can be damaged by inactivity. Cycling is ideal for training the heart to be stronger which results in less stress of the heart. All the risk factors that lead to a heart attack are reduced and regular cycling reduces the likelihood of heart attack by more than 50%.

Body weight, adiposity and obesity

Cycling is ideal for targeting these problems as 70% of the body's weight is borne by the saddle, thus enabling people who could not otherwise move easily to exercise to increase their physical fitness and stimulate fat metabolism. Cycling also contributes to weight reduction by burning energy.

Body fat and hypercholesterol

Cycling can train the organism to use up fat reserves and also changes the cholesterol balance, favouring the protective kind over that which is threatening to health. As body weight reduces and cholesterol is optimised, a protective mechanism is enabled by continuing to cycle. Regular exercise during youth is a prevention factor



against excessive body weight in adults.

Blood pressure

Moderate cycling can prevent, or at least reduce, high blood pressure and so help to avoid stroke or damage to the organs. Blood pressure is also reduced by a lower heart rate, which is a result of regular cycling.

Cancers

Regular physical activity has been shown to reduce the risk of colon, breast, prostate and pancreatic cancers, and possibly lung and endometrial cancers too.

Stamina

Cycling is especially good for aerobic exercise as the strain on the body is less than in other endurance sports. Improved stamina reduces tiredness and fatigue and promotes a sense of well-being..

Beauty and attractiveness

Beauty and attractiveness are much linked to body shape and condition. Cycling can positively influence these by controlling body weight and muscular form. Skin also benefits from the metabolic processes that are stimulated. In addition, cycling affects physical feeling which influences the perception of others.

Physical workout

In addition to regular moderate exercise, the body benefits from more demanding activity from time to time, which improves fitness and provides greater diversion from everyday problems. Cycling can provide more intensive activity as easily as by pedalling faster or harder.

Quality of life

Physical activity has a direct effect on well-being and health. Cycling has numerous advantages that can directly affect quality of life, as it provides benefits both physically and emotionally. Regular exercise, taken as an integral part of daily life, is needed to permanently enhance the quality of life.

Secondary benefits

More cycling, especially as an alternative to motor vehicle travel, would bring substantial health benefits for society as a whole due to improved air quality, reduced noise and danger, and greater independence for children.

References

Principal references

[1] Froböse I. Cycling and Health Compendium. Zentrum für Gesundheit der Deutschen Sporthochschule, Köln. 2004.

[2] Cavill N, Davis A. [Cycling and Health: what's the evidence](#). Cycling England, 2007. **External Link**

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[3] Paffenbarger R et al. [Physical activity, all-cause mortality and longevity of college alumni](#). New England Journal of Medicine, vol. 314(10) pp 605-613, 1986. **External Link**

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- [5] Tuxworth W et al. Health, fitness, physical activity and morbidity of middle aged male factory workers. British Journal of Industrial Medicine vol 43. pp 733-753, 1986.
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- [7] British Medical Association. Cycling: towards health and safety. Oxford University Press, 1992.
- [8] Hillman M. Cycling and the promotion of health. PTRC 20th Summer Annual Meeting, Proceedings of Seminar B, pp 25-36, 1992.
- [9] Jacobsen PL. [Safety in numbers: more walkers and bicyclists, safer walking and bicycling](#). Injury Prevention, 2003;9:205-209. **External Link**
- [10] Robinson DL. [Safety in numbers in Australia: more walkers and bicyclists, safer walking and bicycling](#). Health Promotion Journal of Australia, 2005;16:47-51. **External Link**
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The Bicycle Helmet Research Foundation (BHRF), an incorporated body with an international membership, exists to undertake, encourage and spread the scientific study of the use of bicycle helmets. Also to consider the effect of the promotion and use of helmets on the perception of cycling in terms of risk and the achievement of wider public health and societal goals.

BHRF strives to provide a resource of best-available factual information to assist the understanding of a complex subject, and one where some of the reasoning may conflict with received opinion. In particular BHRF seeks to provide access to a wider range of information than is commonly made available by those that take a strong helmet promotion stance. It is hoped that this will assist informed judgements about the pros and cons of cycle helmets.

For more information, please visit www.cyclehelmets.org.

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